

Titel der Veranstaltung

Optimization for Machine Learning (Vertiefungsbereich "Computational Physics" im Physik-Master)

Art der Veranstaltung

Vorlesung 4 SWS + Übung 2 SWS

Name des Dozierenden

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Inhalt (auf Englisch, da der Kurs auf Englisch angeboten wird):

The course presents various existing optimization techniques for such important machine learning tasks, as inference and learning for graphical models and neural networks. In particular, it addresses such topics as combinatorial algorithms, integer linear programs, scalable convex and non-convex optimization and convex duality theory. Graphical models and neural networks play a role of working examples along the course. The goal of the course is to give a strong background for analysis of existing, and development of new scalable optimization techniques for machine learning problems.

Table of Content:

I. Graphical Models

Acyclic Graphical Models. Dynamic Programming
Background: Basics of Linear Programs and Their Geometry
Inference in Graphical Models as Integer Linear Program
Background: Basics of Convex Analysis and Convex Duality
Duality of the LP Relaxation of Inference Problem
Background: Basics of Convex Optimization
Sub-Gradient and Block-Coordinate Ascent for Inference in Graphical Models
Lagrangian (Dual) Decomposition
Min-Cut/Max-Flow Based Inference
LP Relaxation of Inference Problem as st-Min-Cut Problem
Summary: Inference Algorithm Selection

II Neural Networks

Overview of Architectures
Backpropagation Algorithm
Background: Stochastic Gradient Optimization and its Variants
Stochastic Gradient for Training Neural Networks

III Structured Learning of Graphical Models

Structured Risk Minimization for Graphical Models
CRF+CNN Models: Joint Training of Graphical Models and Neural Networks.

Webseite eines ähnlichen Kurses, dass an der TU Dresden gehalten wurde:

<http://cvlab-dresden.de/courses/opt-machine-learning-ss-16/>